

LAB NO.: 6
DISK SCHEDULING ALGORITHM

Lab Exercises:

1. Write a C/C++ Program to simulate the following algorithms find the total no. of cylinder movements for various input requests

(i) FCFS (ii) SSTF (iii) SCAN (iv) C-SCAN

Program:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
void sort(int arr[], int n) {  
    int i, j, temp;  
    for(i = 0; i < n - 1; i++) {  
        for(j = 0; j < n - i - 1; j++) {  
            if(arr[j] > arr[j + 1]) {  
                temp = arr[j];  
                arr[j] = arr[j + 1];  
                arr[j + 1] = temp;  
            }  
        }  
    }  
}
```

```
void fcfs(int req[], int n, int head) {  
    int i, total = 0;  
    for(i = 0; i < n; i++) {  
        total += abs(req[i] - head);  
        head = req[i];  
    }  
    printf("FCFS Total Head Movement = %d\n", total);  
}
```

```
void sstf(int req[], int n, int head) {  
    int visited[50] = {0};  
    int i, count = 0, total = 0;  
  
    while(count < n) {  
        int min = 10000, index = -1;  
        for(i = 0; i < n; i++) {  
            if(!visited[i]) {  
                int dist = abs(req[i] - head);  
                if(dist < min) {  
                    min = dist;  
                    index = i;  
                }  
            }  
        }  
        visited[index] = 1;
```

```

        total += min;
        head = req[index];
        count++;
    }
    printf("SSTF Total Head Movement = %d\n", total);
}

```

```

void scan(int req[], int n, int head, int disk_size) {
    int i, total = 0;
    int arr[50];

    for(i = 0; i < n; i++)
        arr[i] = req[i];

    arr[n] = disk_size - 1;
    n++;

    sort(arr, n);

    int pos;
    for(i = 0; i < n; i++) {
        if(head < arr[i]) {
            pos = i;
            break;
        }
    }

    for(i = pos; i < n; i++) {
        total += abs(arr[i] - head);
        head = arr[i];
    }

    for(i = pos - 1; i >= 0; i--) {
        total += abs(arr[i] - head);
        head = arr[i];
    }

    printf("SCAN Total Head Movement = %d\n", total);
}

```

```

void cscan(int req[], int n, int head, int disk_size) {
    int i, total = 0;
    int arr[50];

    for(i = 0; i < n; i++)
        arr[i] = req[i];

    arr[n] = disk_size - 1;
    arr[n + 1] = 0;
    n += 2;

    sort(arr, n);

```

```

int pos;
for(i = 0; i < n; i++) {
    if(head < arr[i]) {
        pos = i;
        break;
    }
}

for(i = pos; i < n; i++) {
    total += abs(arr[i] - head);
    head = arr[i];
}

for(i = 0; i < pos; i++) {
    total += abs(arr[i] - head);
    head = arr[i];
}

printf("C-SCAN Total Head Movement = %d\n", total);
}

int main() {
    int n, i, head, disk_size;

    printf("Enter number of requests: ");
    scanf("%d", &n);

    int req[50];

    printf("Enter request sequence:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &req[i]);

    printf("Enter initial head position: ");
    scanf("%d", &head);

    printf("Enter disk size: ");
    scanf("%d", &disk_size);

    fcfs(req, n, head);
    sstf(req, n, head);
    scan(req, n, head, disk_size);
    cscan(req, n, head, disk_size);

    return 0;
}

```

Output:

```
Enter number of requests: 12
Enter request sequence:
176 79 34 60 92 11 41 114 151 9 70 180
Enter initial head position: 50
Enter disk size: 200
FCFS Total Head Movement = 860
SSTF Total Head Movement = 212
SCAN Total Head Movement = 339
C-SCAN Total Head Movement = 389
```